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INDUSTRIAL APPLICATIONS OF PULSED-LASER ESPI VIBRATION ANALYSIS

H. Van der Auweraer*[a], H. Steinbichler[b], C. Haberstok, R. Freymann[c], D. Storer[d], V. Linet [e]

[a]LMS International, Interleuvenlaan 68, 3001 Leuven, Belgium
[b]Labor dr. Steinbichler, Am Bauhof 4, 83115 Neubeuern, Germany
[c]BMW AG, Knorrstrasse 147, 80788 München, Germany
[d]Centro Ricerche Fiat, Strada Torino 50, 10043 Orbassano (TO), Italy
[e]PSA, Route de Gisy, 78140 Vélizy, France

ABSTRACT

The present paper reviews the critical elements of a vibration testing system based on pulsed-laser holographic ESPI measurements. Next to the optical parts, the integration with the modal testing and analysis procedure is reviewed. Specific extensions have been added in view of the industrial application of the approach. These include for example the integration of geometry and response information, the integration of multiple views into one single model, the integration with FE model data and the prior identification of the critical panels and critical modes. A global procedure was hence established. The approach has been applied to several industrial case studies, including car panels, the firewall of a monovolume car, a full vehicle, panels of a light truck and a household product.

1. ESPI-HOLOMODAL ANALYSIS

1.1 Introduction

Experimental modal analysis (EMA) is a well-established approach to analyse and model the dynamic behaviour of mechanical structures. The upper frequency where EMA can be applied depends on the type of the structure. The limiting factors with increasing frequency are:

- the increasing spatial complexity;
- the increasing modal density/overlap of the modes.

For light-weight, panel-type structures, the accelerometer weight influences the structural response and non-contact measurements are needed. The most widely spread approach uses a laser-Doppler vibrometer, equipped with a computer controlled position/orientation system.

A second non-contact testing approach is based on holography. Its main advantage is that it allows whole-field measurements of a vibrating structure. Recent evolutions in the field of ESPI (Electronic Speckle Pattern Interferometry) lead to direct CCD-camera imaging of the vibrations. However, so far, ESPI holography was only used to qualitatively "image" vibration patterns, for example at resonance frequencies. In the present research programme, this was extended to derive quantitative data, in the form of whole-field Frequency Response Function (FRF) maps. These data can then be processed by modal parameter estimation methods or they can directly be used for acoustic radiation or FE updating applications.

1.2 ESPI-Holography FRF Measurement System

The principles of Electronic Speckle Pattern Interferometry are well documented [1,2]. For quantitative vibration analysis, first, specklegrams are obtained for well-defined deformation states (for example using stroboscopic CW illumination, or pulsed lasers, triggered to the excitation force). The optical phase is extracted using time-stepping [3,4] or a novel spatial phase difference approach [5].

When the phase fields for two deformation states are known (e.g. one in rest, one a vibration snapshot), these can be subtracted, resulting in contrast-rich fringe patterns. The phase is unwrapped and scaled to relative displacement.

In classical ESPI, measurements are only performed at frequencies of maximal vibration, using the vibration responses as mode shapes [6,7]. For the HOLOMODAL purposes, the goal was to obtain complex-valued vibration amplitudes (at pixel resolution) that, in reference to the excitation force, yield an "image" of frequency response functions (FRFs). These FRF images are accumulated over a range of frequencies [8,9,10]. Stepped-sine excitation technique is used hereto. To determine the unknown value of the complex amplitude, at least two measurements per frequency are needed [11]. When more than two time points are used, the complex amplitude is calculated by a least squares procedure, reducing the effects of measurement data noise and filtering harmonic disturbances in the excitation and response signals.

In practice, a hybrid approach using a preliminary broadband test with a few accelerometers, combined with a

fine-resolution stepped sine sweep around the frequencies of high dynamic response, turned out to be very effective. It combines the speed of the direct response approach with the accuracy of an FRF-based modal parameter estimation.

Two set-ups were realised, one using a stroboscopic CW laser and one using a double pulse laser. The complete set-up of laser, optics, camera and image processing software was developed by Labor Dr. Steinbichler (figure 1). For acquiring the double-pulse images, a special measurement head was developed, allowing the subsequent 12-bit storage of 2 full frames of 1280X1024 specklegram pixels for a minimal pulse separation of 0.5 μs.

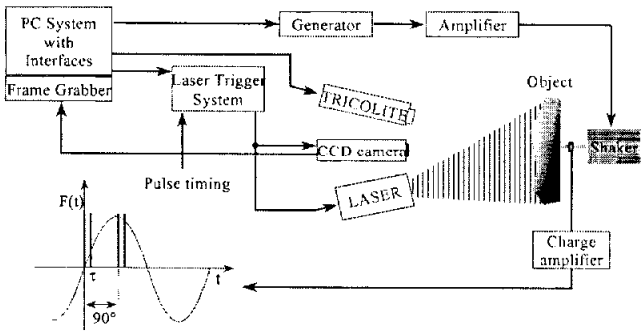

Figure 1: Pulsed-laser ESPI set-up

Also the next evaluation step, filtering and unwrapping the phase maps, was optimised and automated, taking into account information about the quality of the phase values (Figure 2).

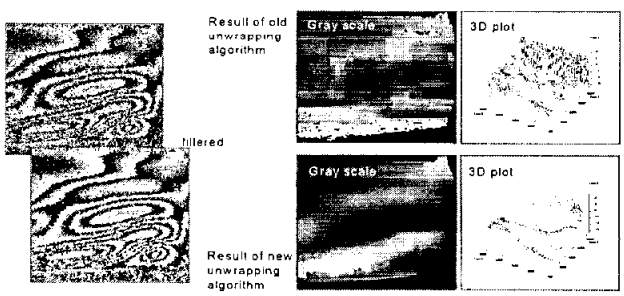

Figure 2: Phase unwrapping results (upper: old method, lower: new method)

After data reduction, the data are transferred to the LMS software for Modal Analysis and CAE applications. This interface is referred to as the "HOLOMODAL Explorer".
A detailed discussion of the developed hardware/software system, as well as its extension to 3-D measurements can be found in [11].

1.3 Modal Parameter Estimation

The identification of modal parameters from ESPI-FRF measurements has to be robust with respect to non-uniform data quality (e.g. due to poor reflecting areas), or unequal vibration levels (due to local modes). Furthermore it must deal with the following requirements:

- to deal with an extremely large number of DOFs;
- to treat high order systems;
- to treat non-uniform FRF data.

The data for estimating the modal parameters are FRFs with image resolution. Since the number of effective degrees of freedom is only a fraction of the number of measured responses, a data reduction procedure is used to compress the information and to reduce the parameter estimation time.

A straightforward way is to subsample the image. However, no use is then made of the redundancy in the data. The selected approach is based on a two-dimensional Spline regression [12,13]. The determination of the Spline basis is done once using the measurement near the highest resonance. The use of Splines as regressive functions increases the computational speed. Also, noise information can be estimated [13]. The result is that, next to the "virtual" FRF matrix, also a matrix with uncertainty information is available (Figure 3).

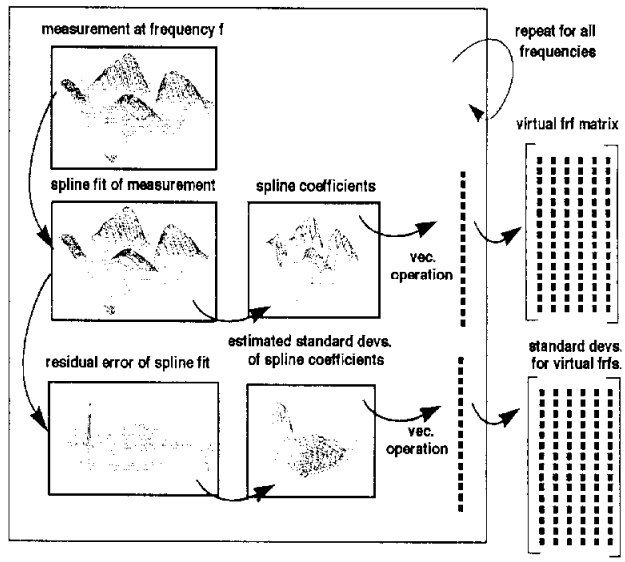

Figure 3: Data reduction procedure

Once the data are condensed, the actual parameter estimation is performed. Since the transformation is linear, the spline coefficients can be used as "virtual measurements" for estimating the modal parameters (poles and "virtual" mode shapes). To handle medium frequency, high-order, models, a weighted frequency domain Maximum Likelihood algorithm was developed, using a fast and robust logarithmic implementation[14]. An essential element is that noise information is explicitly used in the cost function, dramatically improving the estimation process. A set of corresponding uncertainties complements the resulting parameters. Finally, the "virtual mode shapes" can be expanded back into the full measurement grid.

1.4 Geometry and CAE Integration

Critical to using ESPI results in structural dynamics is their reference to a geometry. The object geometry is measured using a LCD fringe projector (Steinbichler TRICOLITE system) projecting different gratings on the object. The

same camera as used for the ESPI views the illuminated object. The co-ordinates of the geometry are calculated by triangulation, using the light code and the CCD camera pixel position. Each pixel of the ESPI image is given a co-ordinate. The geometry information is also transferred to the LMS software.

A specific problem is the relation of this geometry with the description available in a FE model. One solution is to correlate all points of the holographic image to interpolation functions of the FE model. This uses all ESPI information, but it requires access to the FE interpolation functions. But, the image processing corresponds to a data reduction, it is unnecessary to transfer all data. Therefor, a preselection of the available holographic results by defining which of these points correspond to the FE model is also supported. An overview of the related data entities in the HOLOMODAL Explorer is given in figure 4.

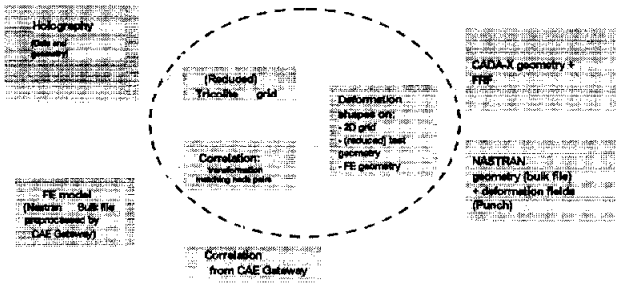

Figure 4: HoloModal CAE Integration

Once the experimental holographic data are available in a CAE reference frame (projected on a FE raster), the data can be used in many CAE applications such as correlation and updating of FE models. In many cases, the main purpose of the laser holography is the validation of panel FE models. The measured FRF data can also be used as boundary conditions to vibro-acoustic prediction codes, for example in acoustic BE or FE codes.

1.5 Multipanel measurements

An essential requirement for vibro-acoustic testing is that measurements inside a cavity can be done. Hereto the holographic recording device was miniaturised and a mirror device for the recording and the illumination system was developed and tested.

Multiple panels are tested one after the other. Panels with complex geometry often are subdivided in multiple parts, measured from different viewpoints. For each position, the geometry and the FRF of the field of view (fov) is measured in a co-ordinate system which is related to the actual position of the measurement head. Upon measurement and acquisition of the panel data, the various sub-images are merged into a global representation of the multi panel structure. This requires to combine several measurements or to image data onto an existing geometry like a FEM mesh. To determine the transformation matrix, the Polyworks software is applied. Different strategies can then be used to combine the different views:

- For each view, 3 or 4 reference points are measured with conventional techniques. Using these reference points, different fields of view can be merged
- Matching of adjacent sub-images using the structure of the surface in the overlapping areas of different fields of view.
- Using photogrammetry and markers on the surface, the multi-panel geometry for the reference points (markers) can be determined. 3 reference points are required for each sub-image.
- The sub-image is linked to a FE Model. This requires a choice of the pixels corresponding to the FEM modes.

1.6 Acoustic excitation

Modal tests mainly use structural excitation. But the shaker may introduce constraints, changing the structure dynamics. This influence will be larger with lighter objects. Therefore an approach for contactless, acoustic, excitation was developed. The formulations are described in [15]. For the excitation, a calibrated volume velocity source is used. Case studies proved that this concept of vibro-acoustical modal analysis is valid. The mode shape, eigenfrequency and damping can be extracted accurately. But a specific problem was found in obtaining correctly scaled modes.

To address this, a procedure was proposed requiring one additional structural driving point measurement (e.g. using hammer excitation). The modal mass is then calculated with the information coming from the structural driving point FRF.

Figure 5 shows the results for a plate modal analysis. A synthesis calculation for an unmeasured structural FRF is compared with an additional measurement. The FRF was one for a reference not used to calculate the new participation factor.

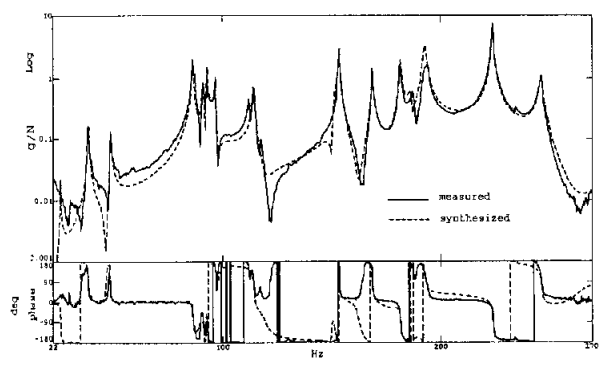

Figure 5: FRF expansion to another reference

It can be concluded that acoustic excitation together with one structural FRF leads to good agreement.

2. APPLICATION TO SINGLE PANELS

The ESPI-HOLOMODAL approach was validated at BMW on a ribbed panel, representative for typical automotive panels such as a cabin or trunk floor plate or a door panel. ESPI-based FRF tests using CW-laser illumination, followed

by modal analysis, have been performed. Also, validations have been performed with respect to the use of ESPI results for acoustic radiation (BEM) predictions. The plate was mounted hereto over a closed cavity and excited by a shaker. These results are discussed elsewhere [16].

2.1 ESPI-LVD comparison

Another validation test at BMW was performed comparing the ESPI system with a scanning laser-Doppler vibrometer (LDV). To correlate the panel with a previously created FE model, several reference points were chosen. During the actual LDV test, the laser beam was guided to every point of the FE model, measuring the FRF from 0 to 800Hz at 1432 points.

The ESPI measurement head was mounted above the test panel. The TRICOLITE system was used to measure the geometry of the panel and to get a correlation of the measured geometry to the camera pixels. The ESPI test was executed from 60 to 460 Hz with a frequency step of 0.5 Hz. After applying a grid, the number of points was reduced to 934.

The frequency response functions, measured with the LDV and the ESPI system are compared for the same measurement points (figure 6). The ESPI measurements are a bit noisier, which is especially the case for low response points. The amplitude is good, but there are some differences in the phase (in addition to a 180° shift). It should be noted that these are raw data, without any spatial smoothing.

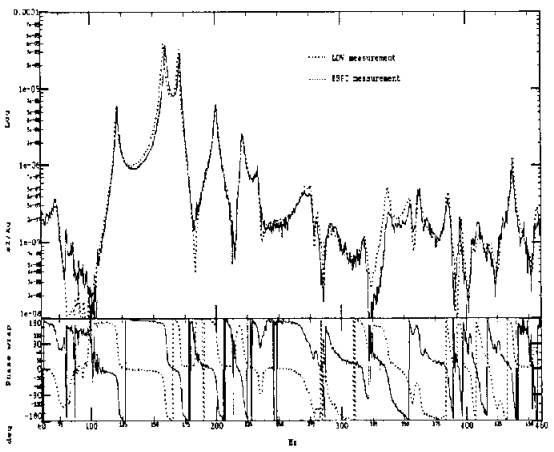

Figure 6: FRF of a response point measured with ESPI and LDV (dotted line)

A modal analysis was then performed on both data sets between 60 and 470 Hz using the LSCE method. In both cases the stabilisation is very good. The same modes are found back in both data sets. For some modes there is a little frequency shift, probably due to a little change of the measurement set-up between the two tests (some weeks apart). The damping estimates are the same for both analyses. The mode shapes are very much the same, as can be seen in figures 7 and 8 for 1 mode. The differences in the generalised modal parameters (unity maximum scaling) are somewhat larger.

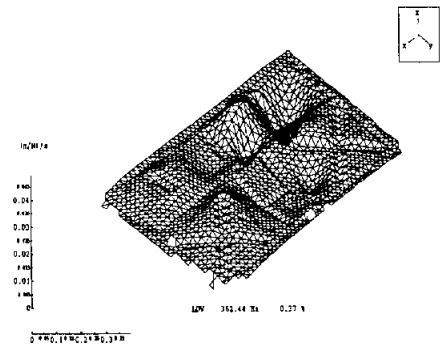

Figure 7: Mode at 362.44 Hz (LDV)

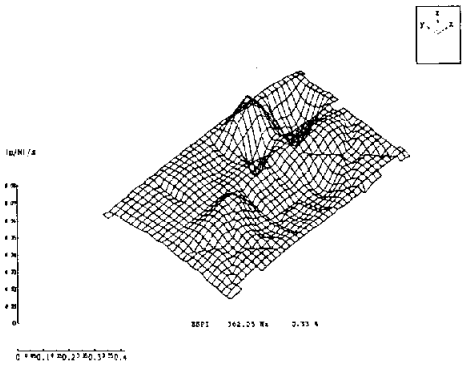

Figure 8: Mode at 362.05 Hz (ESPI)

Overall, it can be concluded that these validations proved the inherent capabilities of the system. In view of data quality, the use of multiple sampling steps per frequency is advised. In view of the measurement time, the laser cooling and recharge time are the dominant ones.

3. MULTIPANEL EVALUATION ON A DASHBOARD

The methodology for merging the data from different panel measurements into one single consistent model was further evaluated by PSA on a Peugeot 806 firewall (Figure 9).

Nine views of 600 x 500 mm were used to cover the whole surface. The 9 measured panel geometries were matched together. The final geometry consisted of 9 000 000 points and was reduced to 900 points. The global accuracy was estimated to be about 0.5mm.

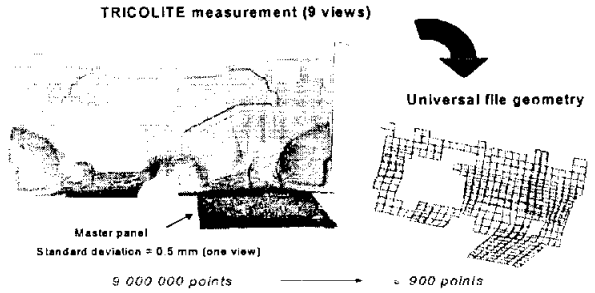

Figure 9: Result of the 9 views matching and transformation into universal file geometry

The comparison of the FRFs in the overlapping regions proved satisfactory. Figure 10 shows the imaginary part of the iso-levels of measured displacements projected on the integrated geometry at one specific frequency (328 Hz).

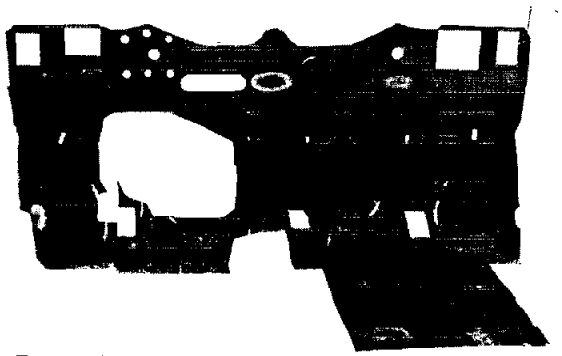

Figure 10: Imaginary part of the sine stepping acquisition at 328Hz

4. FULL VEHICLE STUDY

As a global validation task in the SALOME project, a complete, partially trimmed, vehicle was analysed at CRF with respect to panel induced interior noise. The tested vehicle was a Fiat Punto, 5-doors (old version). In a first step, a panel contribution analysis was performed to identify the most critical panels. The tests were executed on a chassis dynamometer. The setup is shown in Figure 11.

Figure 11: Experimental setup for the contribution analysis

Figures 12 and 13 illustrate the main results for the windscreen, the rear hatch window, and the combined total of the left and right side windows. The graphs represent the 2nd order sound pressure after converting the 1000-5000RPM range into the equivalent frequencies in Hertz.

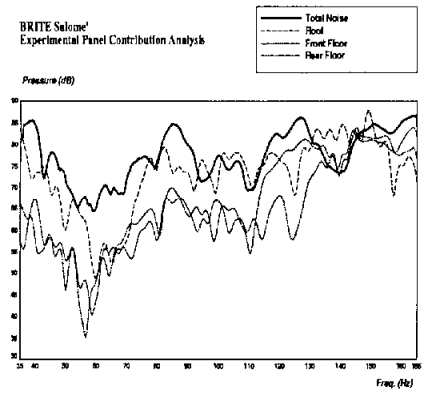

Figure 12: Contributions of the roof and front and rear floor panels

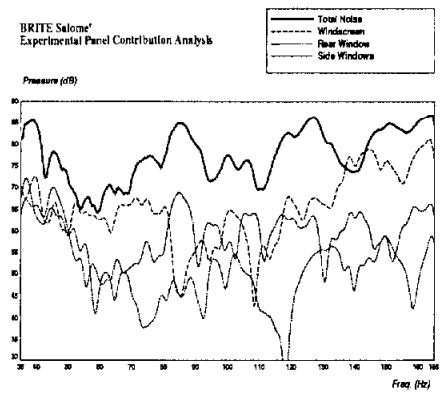

Figure 13: Contributions of the windscreen, the rear hatch window, and the combined total of the left and right side windows

In general it can be observed that for the partially-trimmed vehicle under investigation, the main contribution to the internal sound pressure level is accounted for by the roof and floor panels over the 75-165Hz frequency range, whereas the windscreen has relevant contributions particularly over the 50-70Hz and 135-165Hz ranges. These results correspond to earlier contribution analysis studies conducted on a Body-in-White of the same car type. In these tests, the BIW was closed by adding the windows and doors [17, 18]. Similarly, contribution analysis studies based on numerical and hybrid (mixed numerical/experimental) models yielded comparable results [19, 20].

In a second step, a detailed dynamic analysis of the windscreen was performed using the ESPI system. Stepped sine shaker excitation (40-160 Hz) was applied hereto at the front engine mount. Figure 14 shows the operational deflection shapes of the windscreen at 160 Hz (where the windshield also had a significant contribution).

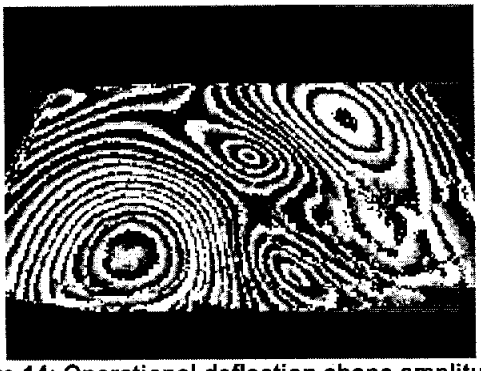

Figure 14: Operational deflection shape amplitude at 160 Hz

Furthermore, a detailed FE model of the trimmed body was made by CRF [19]. In order to validate this model, a comparison of modeshapes was conducted. In Figure 15, the experimentally obtained vibration shape at 100 Hz is compared to the corresponding FE-shape.

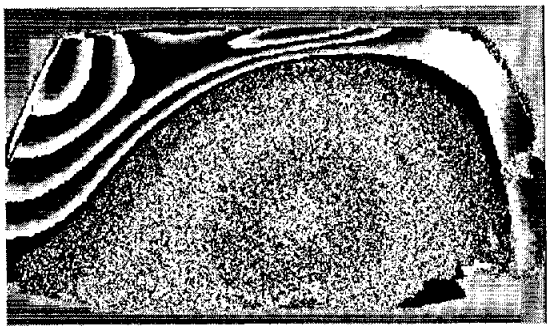

Figure 15a: The ESPI operational deflection shape of the windscreen at the frequency of 100 Hz

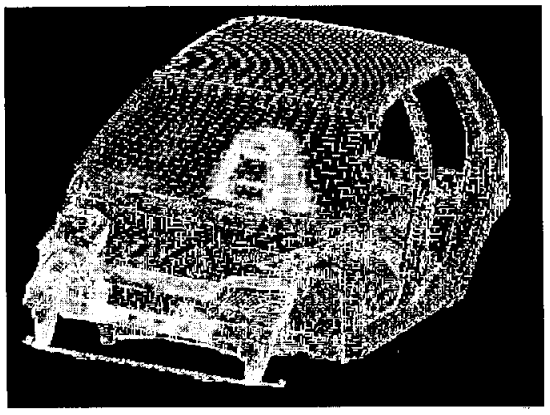

Figure 15b: The (FE-calculated) operational deflection shape of the windscreen at 100Hz

The correspondence between the results is very good. These results also show the importance to first execute a panel –and mode- contribution analysis to identify the panels responsible for the interior noise. The detailed ESPI analysis can then be limited to those critical panels and frequency ranges.

5. APPLICATION TO A HOUSEHOLD PRODUCT

In order to validate the general applicability of the developed system, an analysis was performed on a refrigerator. The measurement was done for one side plate of the housing. The side plates of the housing are excited while the compressor is in operation causing an annoying noise.

Stepped sine measurement was performed between 75 Hz to 81 Hz in steps of 0.3 Hz. Two separate (overlapping) panels were measured. The resulting vibration amplitude response field at 79.4 Hz, overlaid on the geometry, is shown in Figure 16. The vibration amplitude images acquired for the upper and the lower panel fit very well in the overlapping region.

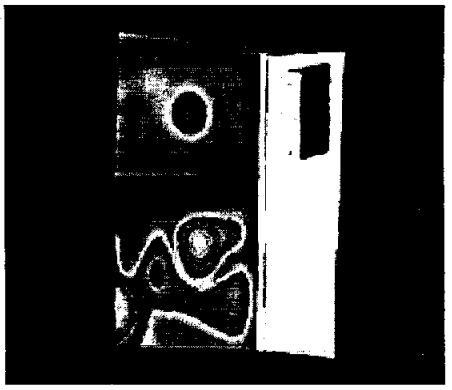

Figure 16: Geometry with overlayed vibration image at 79.8 Hz

The measurement data from the panel (geometry and response data) were imported in the LMS system to perform a modal analysis. One main mode was identified in the tested frequency range, using the data of both panels simultaneously (Figure 17).

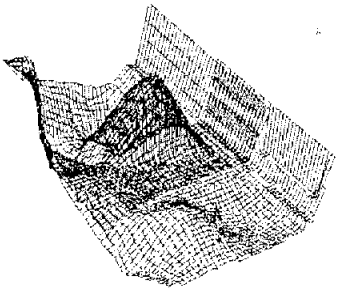

Figure 17: Principal modeshape

6. CONCLUSIONS

The project has been successfully concluded, resulting in significant advances both on the level of modal testing and on the level of vibro-acoustic analysis.

Both levels are very important and complementary: The vibro-acoustic analysis approach allows to identify the critical panels. The testing methodology and tools allow to establish the operational vibration characteristics of the panels as well as to derive accurate modal models. The vibro-acoustic analysis approach then can use such models for designing appropriate design improvements.

At BMW, PSA and CRF, several test structures were realised and an extensive set of validation measurements was performed, leading to various system improvements. The developed methods, tools and procedures have been validated in various industrial problems, including a full car, some car and truck components and a household product.

The technology is proven; the further development to exploitable industrial tools and solutions is ongoing. The involved car manufacturers have acquired the background and know-how on the new methodologies and are ready to exploit this in their daily business.

7. ACKNOWLEDGEMENTS

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